ALGEBRA 1 · UNIT 1

Section 1.3 Function Families

Function families are groups of relationships that behave in similar ways. The skill is using evidence, not just appearance, to recognize the behavior.

> **Mathematical idea:** A function family is not just a graph shape to memorize. It is a pattern of change that can appear in tables, equations, graphs, and contexts.

Today we will practice identifying common function families, comparing how they behave, and defending classifications with evidence from more than one representation.

Learning Target

Students will recognize common function families from graphs, tables, equations, and contexts.

I can statements

- I can identify common function families from graphs.
- I can identify common function families from equations.
- I can describe how function families behave.
- I can compare different function families.
- I can distinguish classification from full modeling.

What's To Come

This is a long-range preview for future lessons. Do not solve it today. Instead, annotate it individually. Circle anything familiar, underline symbols or directions that seem important, and write notes about what information you think will be needed later.

As you annotate, ask yourself: Which representation would you trust first? Which representation might be misleading? What extra evidence would help you classify the relationship?

Future Task — Conflicting Family Evidence

A new school app tracks total downloads after students begin sharing it. The data below shows only the first three weeks. Different students disagree about the function family.

Graph

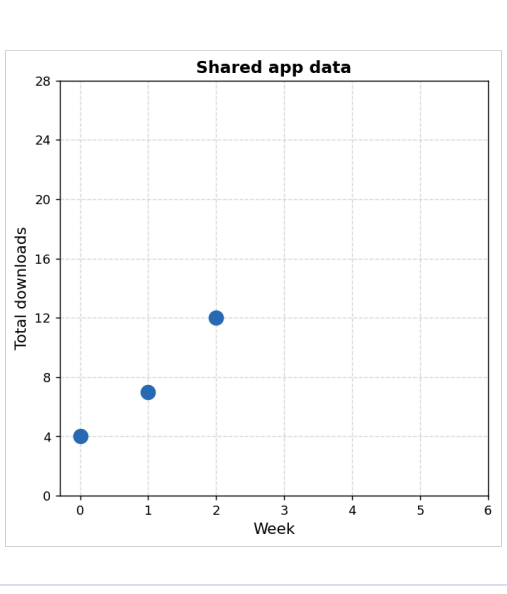


Table (weeks 0–2 only)

Week (w)	0	1	2
Downloads (D)	4	7	12

Student A

"It is probably linear because the points mostly move upward in a steady direction."

Student B

"It might not be linear because the number of new downloads seems to be changing — but I need more data to be sure."

1. Circle one piece of evidence from the table or graph that seems useful for classification.
2. Underline one piece of evidence that might be misleading with so little data.
3. Both students agree more data is needed. Write two specific input-output pairs you would want — and explain what pattern in those values would support each student's position.

Intro

Directions

Read the introduction aloud with a partner or in a small group. As you read, annotate the text by highlighting important ideas, circling key vocabulary, and writing short notes about connections, questions, or predictions in the margins.

A function family is a group of functions that share similar behavior. Members of the same family may have different numbers in their equations, but they change in a similar way.

For example, a linear relationship changes by adding the same amount again and again. A quadratic relationship often grows by changing amounts that form a pattern. An absolute value relationship has a turning point and two straight branches. An exponential relationship changes by multiplying by a common factor.

Graphs can help us notice families, but graph shape alone is not enough. A rough sketch, a small window, or a few points can be misleading. Tables, equations, and contexts give additional evidence about how the output changes as the input changes.

In this section, classification means making a reasonable family decision from evidence. Classification is not the same as full modeling. A classification tells us the likely type of behavior; a full model gives a specific rule that matches the situation.

Stop and Discuss

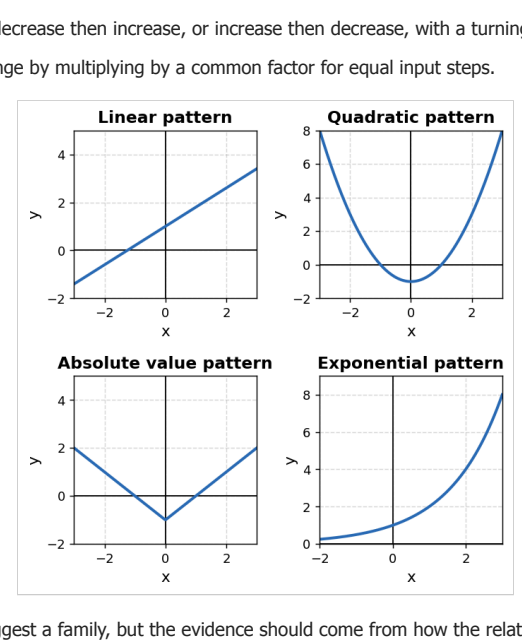
1. According to the reading, why is graph shape alone not enough evidence?
2. What is the difference between classifying a function family and building a full model?
3. Which representation do you think would be most useful for checking how outputs change: a graph, table, equation, or context? Defend your choice.

Families Are Patterns of Behavior

MATH NOTES **Common Function Families**

Each function family has a typical way that outputs change.

- **Linear:** outputs change by adding the same amount for equal input steps.
- **Quadratic:** outputs often curve because the changes themselves change in a regular way.
- **Absolute value:** outputs decrease then increase, or increase then decrease, with a turning point.
- **Exponential:** outputs change by multiplying by a common factor for equal input steps.



Important: The graph can suggest a family, but the evidence should come from how the relationship behaves.

YOUR NOTES

Example 1: Analyze a Table for Family Behavior

A student wants to classify this relationship.

Input (x)	0	1	2	3	4
Output (y)	6	10	14	18	22

1. Find how the output changes each time the input increases by 1.
2. Which family is best supported by this table: linear, quadratic, absolute value, or exponential?
3. Write one sentence explaining why your classification is based on behavior, not just a guess.

You Try It 1: Compare Table Evidence

Two tables are shown. One student says both tables are linear because both outputs increase.

Table A

x	0	1	2	3
y	3	6	9	12

Table B

x	0	1	2	3
y	3	6	12	24

1. For each table, describe how the outputs change.
2. Which table has stronger evidence of linear behavior? Which table has stronger evidence of a different behavior?
3. Explain why "both outputs increase" is not enough evidence to classify the family.

MATH NOTES **Evidence Before Labels**

When classifying a function family, collect evidence before choosing a label.

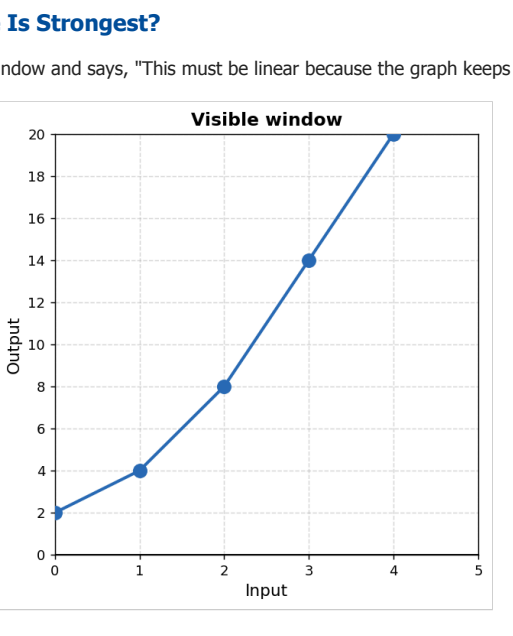
- **Table evidence:** Look at how outputs change for equal input steps.
- **Equation evidence:** Look at how the variable is used: x, x^2, $|x|$, or a variable in an exponent.
- **Graph evidence:** Look for shape, but check that the window and scale are not hiding important behavior.
- **Context evidence:** Look for words that suggest constant adding, changing growth, turning around, or repeated multiplying.

Evidence sentence frame: I think this family is __________ because the __________ representation shows __________.

YOUR NOTES

Example 2: Which Evidence Is Strongest?

A student looks at the visible graph window and says, "This must be linear because the graph keeps going up."



x	0	1	2	3	4
y	2	4	8	14	20

Claim: "Increasing means linear."

1. Use the table to check how the outputs change. Are the changes constant?
2. What evidence does the graph provide? What evidence does it *not* provide by itself?
3. Do you agree, disagree, or need more information? Defend your response using at least two pieces of evidence.

You Try It 2: Defend a Classification

A relationship is represented by the equation $y = x^2 + 2$ and the partial table below.

x	0	1	2	3
y	2	3	6	11

1. What does the equation suggest about the function family?
2. How do the output changes in the table support or challenge that classification?
3. Write a defense of the classification using both the equation and the table. Avoid saying only, "because it looks like a parabola."

Stop and Discuss

A student says, "I do not need a table if I already have a graph." What might the student miss by relying only on the graph?

MATH NOTES **Shape Can Mislead**

Many incorrect classifications come from using one weak clue.

- **Weak clue:** "It goes up." Many families can increase.
- **Weak clue:** "The equation has an x." Many equations contain variables.
- **Weak clue:** "It looks curved." A rough graph or small window can hide behavior.
- **Stronger clue:** Use a table, equation, context, or multiple graph features to explain the behavior.

Revision frame: The claim is incomplete because __________. A stronger claim would use __________ as evidence.

WEAK CLUE

STRONGER EVIDENCE

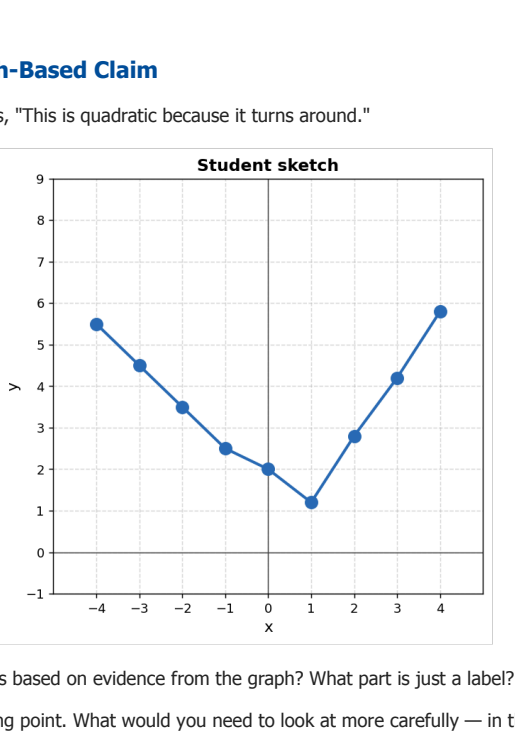
Example 3: Critique a Classification Claim

Student claim: "The equation $y = x^2 + 3$ is linear because it has an x in it."

1. What is the misconception in the student's reasoning?
2. Create a small table for $x = 0$, $x = 1$, $x = 2$, and $x = 3$.
3. Use the equation and your table to write a corrected classification claim.

You Try It 3: Revise a Graph-Based Claim

A student looks at the sketch and says, "This is quadratic because it turns around."



1. What part of the student's claim is based on evidence from the graph? What part is just a label?
2. At least two families have a turning point. What would you need to look at more carefully — in the graph or in a table — to decide between them?
3. Revise the student's claim so it names what specific evidence would need to be true for the quadratic classification to be correct. What would have to be true about the branches?

MATH NOTES **Families in Context**

Contexts can suggest a family, but context clues should still be checked with numbers.

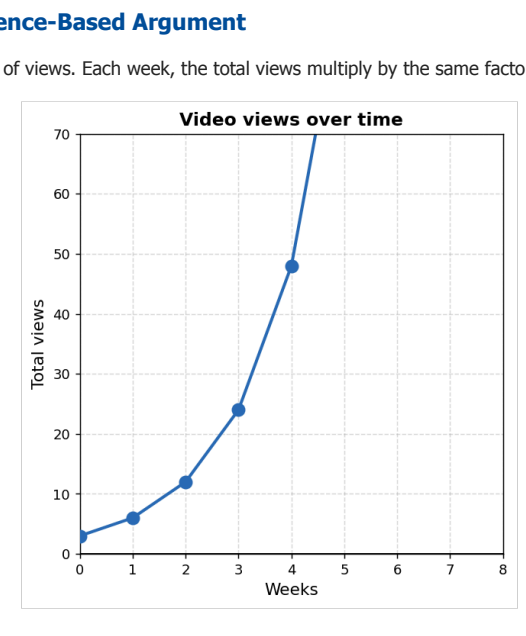
- **Constant adding:** often suggests linear behavior.
- **Area or height-and-width situations:** can suggest quadratic behavior.
- **Distance from a center or turning point:** can suggest absolute value behavior.
- **Repeated doubling, halving, or multiplying:** often suggests exponential behavior.

Classification is a first decision. A full model needs a specific equation, table, graph, and interpretation.

YOUR NOTES

Example 4: Build an Evidence-Based Argument

A video starts with a small number of views. Each week, the total views multiply by the same factor.



Week (w)	0	1	2	3	4
Views (V)	3	6	12	24	48

1. A student looks only at the graph and says, "This is exponential." Another student says the graph is not enough — they need the table too. Who has the stronger position? What does the table add that the graph alone does not tell you?
2. Use the table to show specifically whether the relationship is based on constant adding or repeated multiplying. Show the calculation that proves your answer.
3. What would still be needed to build a full model, rather than only classifying the family?

You Try It 4: Create an Example and a Non-Example

Choose one family: linear, quadratic, absolute value, or exponential.

1. Create a small table that fits your chosen family.
2. Write one context that could reasonably fit your table.
3. Create one non-example that might trick someone at first. Explain exactly why it does *not* fit the family you chose.
4. Write one sentence explaining the difference between classifying your relationship and building a full model for it.

Summary

Function families describe patterns of behavior. A family classification should be based on evidence from how the relationship changes, not only on how a graph looks.

Tables can show whether outputs change by constant adding, changing differences, turning behavior, or repeated multiplying. Equations can reveal structure such as x, x^2, $|x|$, or a variable in the exponent. Contexts can explain why a certain type of change makes sense.

Classification is not the same as full modeling. Classification names the likely family. A full model gives a specific rule and explains how the representations match the situation.

Common Mistakes

- **Matching by graph shape only.** Why it is tempting: the shape is easy to see quickly. What to check instead: use table changes, equation structure, and context clues as evidence.
- **Thinking every increasing graph is linear.** Why it is tempting: the graph moves upward in one direction. What to check instead: ask whether the output changes by the same amount each time.
- **Classifying by the presence of x alone.** Why it is tempting: many equations contain x. What to check instead: look at how x is used: x, x^2, $|x|$, or an exponent.
- **Confusing classification with a full model.** Why it is tempting: naming the family feels like an answer. What to check instead: ask whether you also have a specific equation, table, graph, and context.
- **Trusting a small graph window too much.** Why it is tempting: the visible part looks convincing. What to check instead: use more input-output values or another representation before deciding.

Optional Future Task Reflection

Look back at the future task. Do not solve it yet. Highlight one part that makes more sense now than it did at the beginning. Circle one part that still feels unfamiliar. Write one sentence about what representation you would look at first.